

Knowva ERD

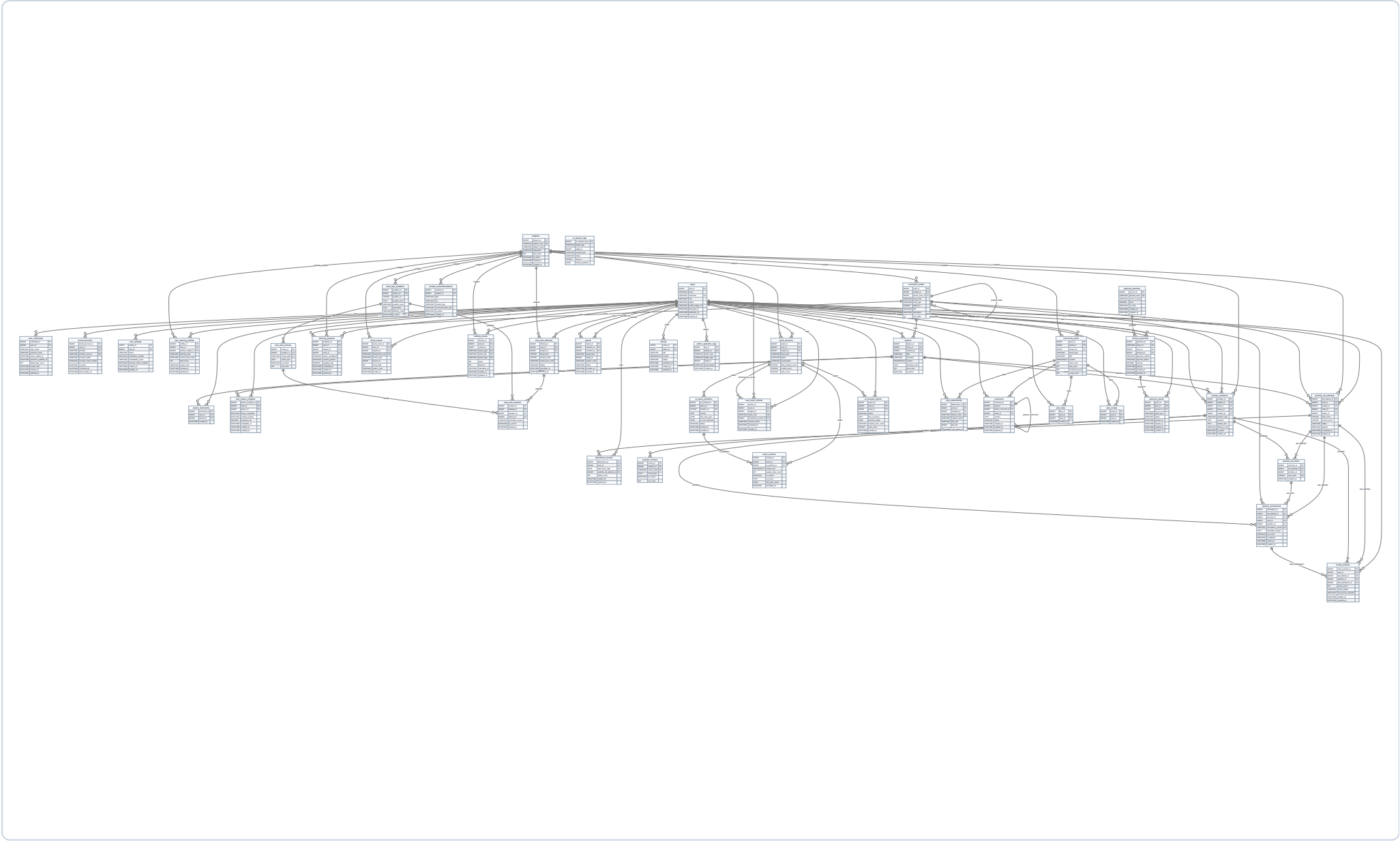
Source: Notion DB 명세 v1.9 Mermaid ERD. 커뮤니티 조회수와 seed 실행 안전성, 인증 수단별 독립 계정 정책을 반영한 HTML 원본.

Version v1.9	Tables 42	Foreign Keys 78	DBMS MySQL 8
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Table	PK	FK Count
users	BIGINT user_id PK	0
subjects	BIGINT subject_id PK	0
curriculum_nodes	BIGINT node_id PK	2
payment_products	BIGINT product_id PK	0
user_credentials	BIGINT credential_id PK	1
social_accounts	BIGINT social_account_id PK	1
user_settings	BIGINT setting_id PK	1
user_learning_profiles	BIGINT profile_id PK	2
lessons	BIGINT lesson_id PK	2
lesson_bookmarks	BIGINT bookmark_id PK	2
user_lesson_progress	BIGINT lesson_progress_id PK	2
level_test_questions	BIGINT question_id PK	1
level_test_choices	BIGINT choice_id PK	1
level_test_attempts	BIGINT attempt_id PK	2
level_test_answers	BIGINT answer_id PK	3
learning_progress	BIGINT progress_id PK	3
practice_set_attempts	BIGINT set_attempt_id PK	4
attendance_records	BIGINT attendance_id PK	2
practice_problems	BIGINT problem_id PK	4
problem_choices	BIGINT choice_id PK	1
practice_set_items	BIGINT set_item_id PK	2
practice_submissions	BIGINT submission_id PK	4
wrong_answers	BIGINT wrong_answer_id PK	4
score_events	BIGINT score_event_id PK	2
ranking_scores	BIGINT ranking_id PK	2
exam_sessions	BIGINT exam_id PK	2
user_level_unlocks	BIGINT unlock_id PK	3
ai_exam_problems	BIGINT ai_problem_id PK	1

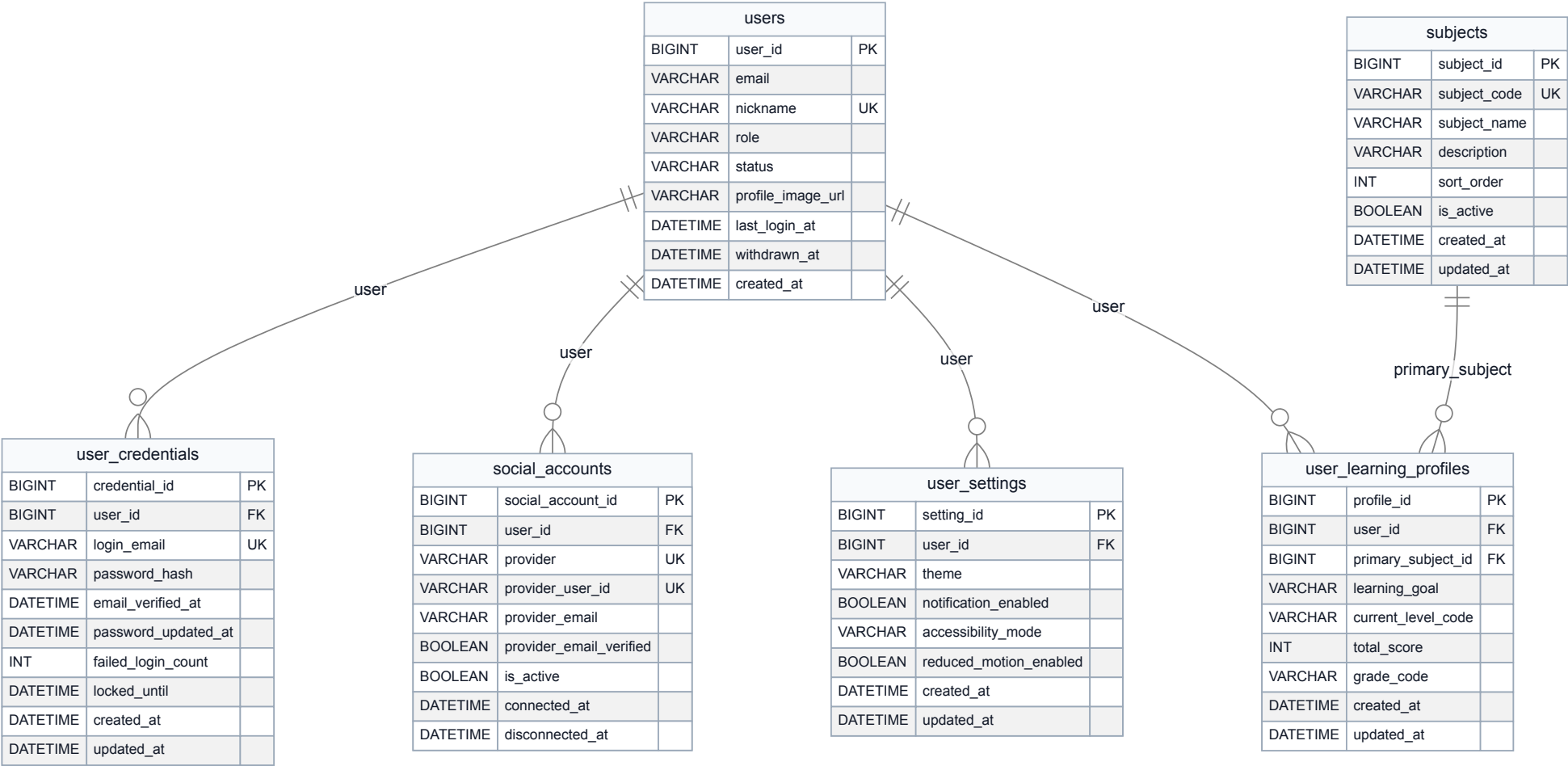
Table	PK	FK Count
exam_answers	BIGINT answer_id PK	2
ai_analysis_reports	BIGINT report_id PK	2
ai_request_logs	BIGINT ai_request_log_id PK	0
community_posts	BIGINT post_id PK	2
post_attachments	BIGINT attachment_id PK	2
comments	BIGINT comment_id PK	3
post_likes	BIGINT like_id PK	2
post_scraps	BIGINT scrap_id PK	2
reports	BIGINT report_id PK	2
dummy_payments	BIGINT payment_id PK	2
premium_grants	BIGINT grant_id PK	2
notices	BIGINT notice_id PK	1
admin_operation_logs	BIGINT log_id PK	1
content_recommendations	BIGINT content_id PK	1

전체 ERD



전체 구조 확인용 축소 ERD. 상세 컬럼과 관계는 이후 섹션별 ERD에서 확인.

회원 / 인증 / 설정



회원 / 인증 / 설정 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
user_credentials	user_id	users	user_id
social_accounts	user_id	users	user_id
user_settings	user_id	users	user_id
user_learning_profiles	user_id	users	user_id
user_learning_profiles	primary_subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL		NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_credentials			
Column	Type	Key	NN
credential_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
login_email	VARCHAR(255) NOT NULL	UK	NN
password_hash	VARCHAR(255) NOT NULL		NN
email_verified_at	DATETIME NOT NULL		NN
password_updated_at	DATETIME NULL		NULL
failed_login_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
locked_until	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

social_accounts			
Column	Type	Key	NN
social_account_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
provider	VARCHAR(30) NOT NULL	UK	NN
provider_user_id	VARCHAR(191) NOT NULL	UK	NN
provider_email	VARCHAR(255) NULL		NULL
provider_email_verified	TINYINT(1) NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
connected_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
disconnected_at	DATETIME NULL		NULL

user_settings

Column	Type	Key	NN
setting_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
theme	VARCHAR(30) NOT NULL DEFAULT 'SYSTEM'		NN
notification_enabled	TINYINT(1) NOT NULL DEFAULT 1		NN
accessibility_mode	VARCHAR(30) NULL		NULL
reduced_motion_enabled	TINYINT(1) NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

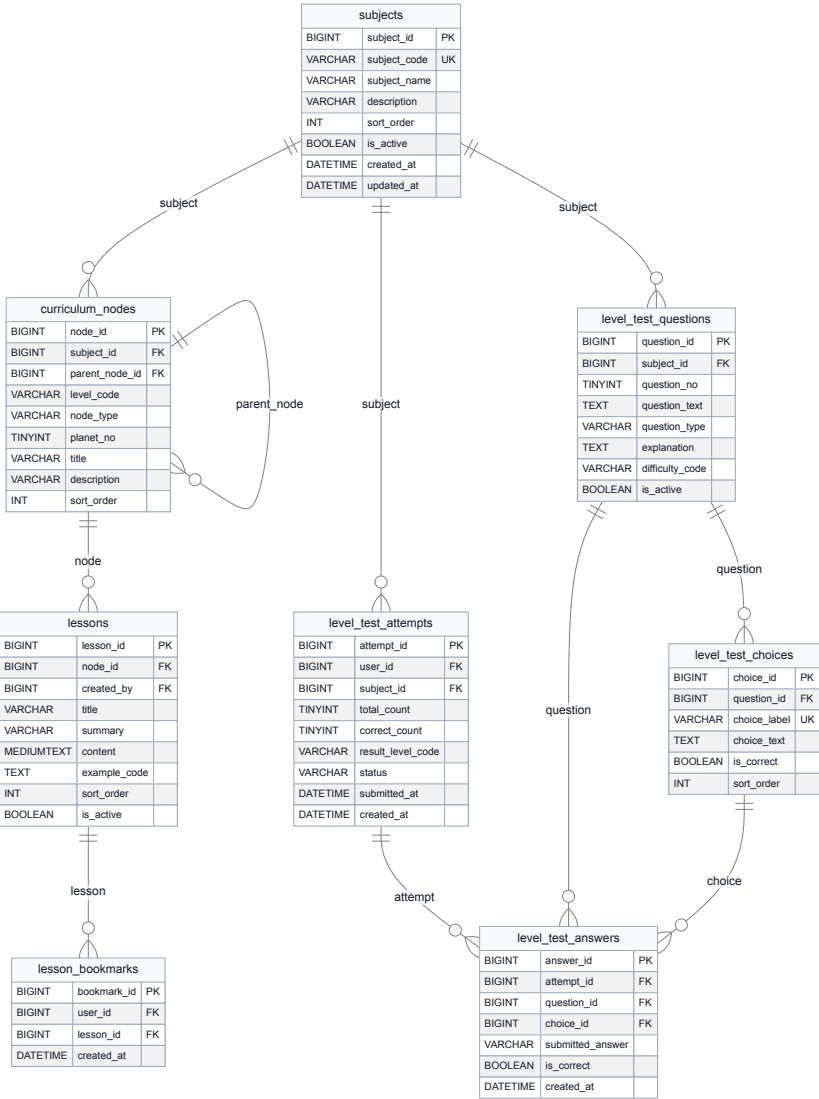
user_learning_profiles

Column	Type	Key	NN
profile_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
primary_subject_id	BIGINT UNSIGNED NULL	FK	NULL
learning_goal	VARCHAR(500) NULL		NULL
current_level_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
total_score	INT UNSIGNED NOT NULL DEFAULT 0		NN
grade_code	VARCHAR(30) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

학습 Master / 레벨 테스트



학습 Master / 레벨 테스트 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
curriculum_nodes	subject_id	subjects	subject_id
curriculum_nodes	parent_node_id	curriculum_nodes	node_id
lessons	node_id	curriculum_nodes	node_id
lesson_bookmarks	lesson_id	lessons	lesson_id
level_test_questions	subject_id	subjects	subject_id
level_test_choices	question_id	level_test_questions	question_id
level_test_attempts	subject_id	subjects	subject_id
level_test_answers	attempt_id	level_test_attempts	attempt_id
level_test_answers	question_id	level_test_questions	question_id
level_test_answers	choice_id	level_test_choices	choice_id

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

curriculum_nodes			
Column	Type	Key	NN
node_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_node_id	BIGINT UNSIGNED NULL	FK	NULL
level_code	VARCHAR(30) NOT NULL		NN
node_type	VARCHAR(30) NOT NULL DEFAULT 'PLANET'		NN
planet_no	TINYINT UNSIGNED NULL		NULL
title	VARCHAR(150) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
gate_condition	VARCHAR(255) NULL		NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

lessons			
Column	Type	Key	NN
lesson_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(150) NOT NULL		NN
summary	VARCHAR(500) NULL		NULL
content	MEDIUMTEXT NOT NULL		NN
example_code	TEXT NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_by	BIGINT UNSIGNED NULL	FK	NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

lesson_bookmarks

Column	Type	Key	NN
bookmark_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

level_test_questions

Column	Type	Key	NN
question_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
question_no	TINYINT UNSIGNED NOT NULL		NN
question_text	TEXT NOT NULL		NN
question_type	VARCHAR(30) NOT NULL DEFAULT 'MULTIPLE_CHOICE'		NN
explanation	TEXT NULL		NU LL
difficulty_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

level_test_choices

Column	Type	Key	NN
choice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
question_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_label	VARCHAR(10) NOT NULL	UK	NN
choice_text	TEXT NOT NULL		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN

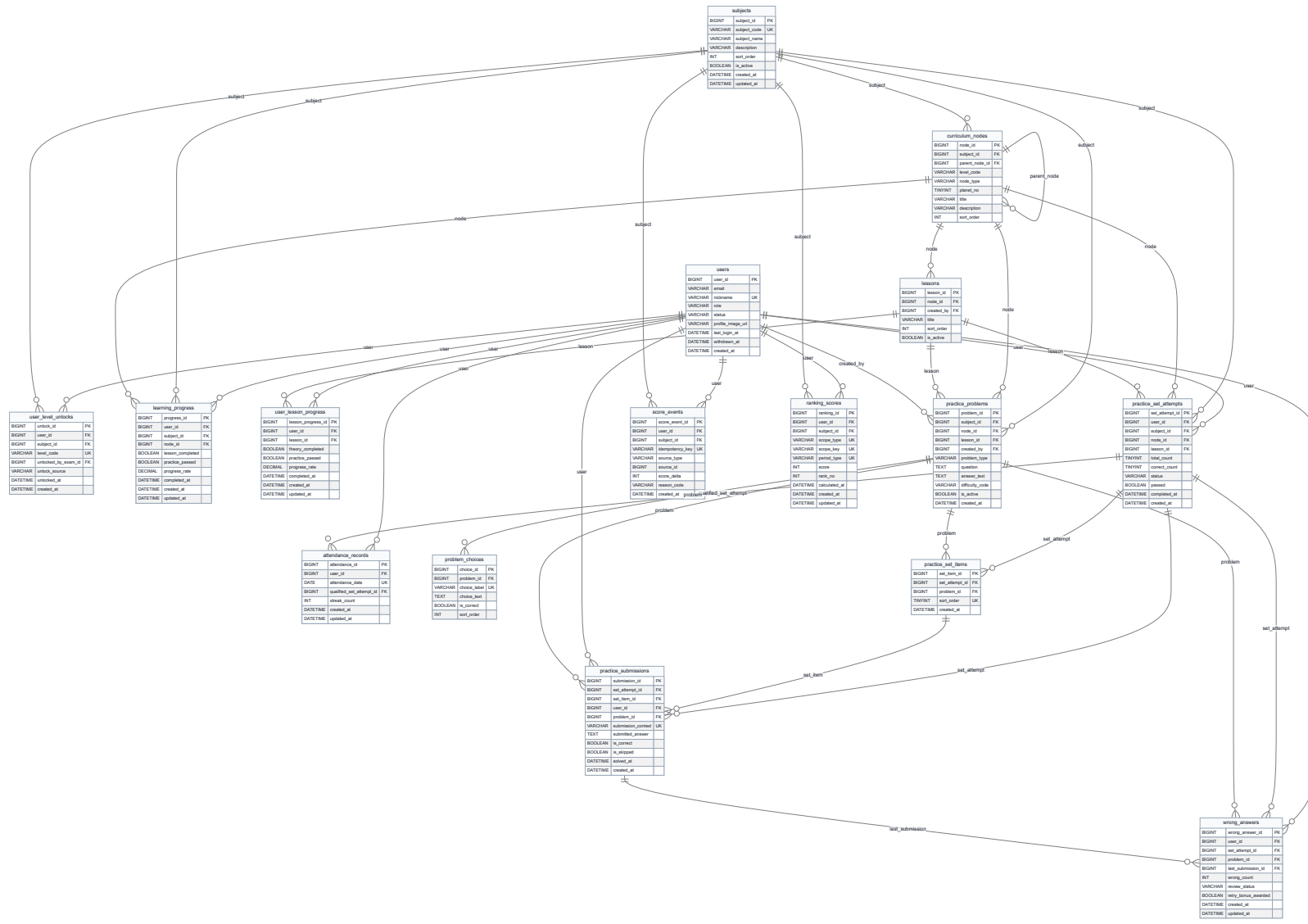
level_test_attempts

Column	Type	Key	NN
attempt_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
total_count	TINYINT UNSIGNED NOT NULL DEFAULT 8		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
result_level_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'SUBMITTED'		NN
submitted_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

level_test_answers

Column	Type	Key	NN
answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
question_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_id	BIGINT UNSIGNED NULL	FK	NU LL
submitted_answer	VARCHAR(500) NULL		NU LL
is_correct	TINYINT(1) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

학습 기록 / 일반 문제 / 점수



Child Table	FK Column	Parent Table	Parent PK
curriculum_nodes	subject_id	subjects	subject_id
curriculum_nodes	parent_node_id	curriculum_nodes	node_id
user_level_unlocks	user_id	users	user_id
user_level_unlocks	subject_id	subjects	subject_id
learning_progress	user_id	users	user_id
learning_progress	subject_id	subjects	subject_id
learning_progress	node_id	curriculum_nodes	node_id
user_lesson_progress	user_id	users	user_id
user_lesson_progress	lesson_id	lessons	lesson_id
attendance_records	user_id	users	user_id
attendance_records	qualified_set_attempt_id	practice_set_attempts	set_attempt_id
practice_set_attempts	user_id	users	user_id
practice_set_attempts	subject_id	subjects	subject_id
practice_set_attempts	node_id	curriculum_nodes	node_id
practice_set_attempts	lesson_id	lessons	lesson_id
practice_problems	subject_id	subjects	subject_id
practice_problems	node_id	curriculum_nodes	node_id
practice_problems	lesson_id	lessons	lesson_id
practice_problems	created_by	users	user_id
problem_choices	problem_id	practice_problems	problem_id
practice_set_items	set_attempt_id	practice_set_attempts	set_attempt_id
practice_set_items	problem_id	practice_problems	problem_id
practice_submissions	set_attempt_id	practice_set_attempts	set_attempt_id
practice_submissions	set_item_id	practice_set_items	set_item_id
practice_submissions	user_id	users	user_id
practice_submissions	problem_id	practice_problems	problem_id
wrong_answers	user_id	users	user_id
wrong_answers	set_attempt_id	practice_set_attempts	set_attempt_id
wrong_answers	problem_id	practice_problems	problem_id
wrong_answers	last_submission_id	practice_submissions	submission_id

Child Table	FK Column	Parent Table	Parent PK
score_events	user_id	users	user_id
score_events	subject_id	subjects	subject_id
ranking_scores	user_id	users	user_id
ranking_scores	subject_id	subjects	subject_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

curriculum_nodes

Column	Type	Key	NN
node_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_node_id	BIGINT UNSIGNED NULL	FK	NULL
level_code	VARCHAR(30) NOT NULL		NN
node_type	VARCHAR(30) NOT NULL DEFAULT 'PLANET'		NN
planet_no	TINYINT UNSIGNED NULL		NULL
title	VARCHAR(150) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
gate_condition	VARCHAR(255) NULL		NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_level_unlocks

Column	Type	Key	NN
unlock_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
level_code	VARCHAR(30) NOT NULL	UK	NN
unlock_source	VARCHAR(30) NOT NULL		NN
unlocked_by_exam_id	BIGINT UNSIGNED NULL	FK	NULL
unlocked_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

learning_progress

Column	Type	Key	NN
progress_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_completed	TINYINT(1) NOT NULL DEFAULT 0		NN
practice_passed	TINYINT(1) NOT NULL DEFAULT 0		NN
progress_rate	DECIMAL(5,2) NOT NULL DEFAULT 0.00		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_lesson_progress

Column	Type	Key	NN
lesson_progress_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
theory_completed	TINYINT(1) NOT NULL DEFAULT 0		NN
practice_passed	TINYINT(1) NOT NULL DEFAULT 0		NN
progress_rate	DECIMAL(5,2) NOT NULL DEFAULT 0.00		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

attendance_records

Column	Type	Key	NN
attendance_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
attendance_date	DATE NOT NULL	UK	NN
streak_count	INT UNSIGNED NOT NULL DEFAULT 1		NN
qualified_set_attempt_id	BIGINT UNSIGNED NULL	FK UK	NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

practice_set_attempts

Column	Type	Key	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
lesson_id	BIGINT UNSIGNED NULL	FK	NULL
total_count	TINYINT UNSIGNED NOT NULL DEFAULT 10		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
status	VARCHAR(30) NOT NULL DEFAULT 'IN_PROGRESS'		NN
passed	TINYINT(1) NOT NULL DEFAULT 0		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

practice_problems

Column	Type	Key	NN
problem_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
lesson_id	BIGINT UNSIGNED NULL	FK	NULL
problem_type	VARCHAR(30) NOT NULL DEFAULT 'MULTIPLE_CHOICE'		NN
question	TEXT NOT NULL		NN
answer_text	TEXT NULL		NULL
difficulty_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
created_by	BIGINT UNSIGNED NULL	FK	NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

problem_choices

Column	Type	Key	NN
choice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_label	VARCHAR(10) NOT NULL	UK	NN
choice_text	TEXT NOT NULL		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN

practice_set_items

Column	Type	Key	NN
set_item_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
sort_order	TINYINT UNSIGNED NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

practice_submissions

Column	Type	Key	NN
submission_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
set_item_id	BIGINT UNSIGNED NULL	FK	NULL
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
submission_context	VARCHAR(30) NOT NULL DEFAULT 'PRACTICE_SET'	UK	NN
submitted_answer	TEXT NULL		NULL
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
is_skipped	TINYINT(1) NOT NULL DEFAULT 0		NN
solved_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

wrong_answers

Column	Type	Key	NN
wrong_answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
last_submission_id	BIGINT UNSIGNED NULL	FK	NULL
wrong_count	INT UNSIGNED NOT NULL DEFAULT 1		NN
review_status	VARCHAR(30) NOT NULL DEFAULT 'OPEN'		NN
retry_bonus_awarded	TINYINT(1) NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

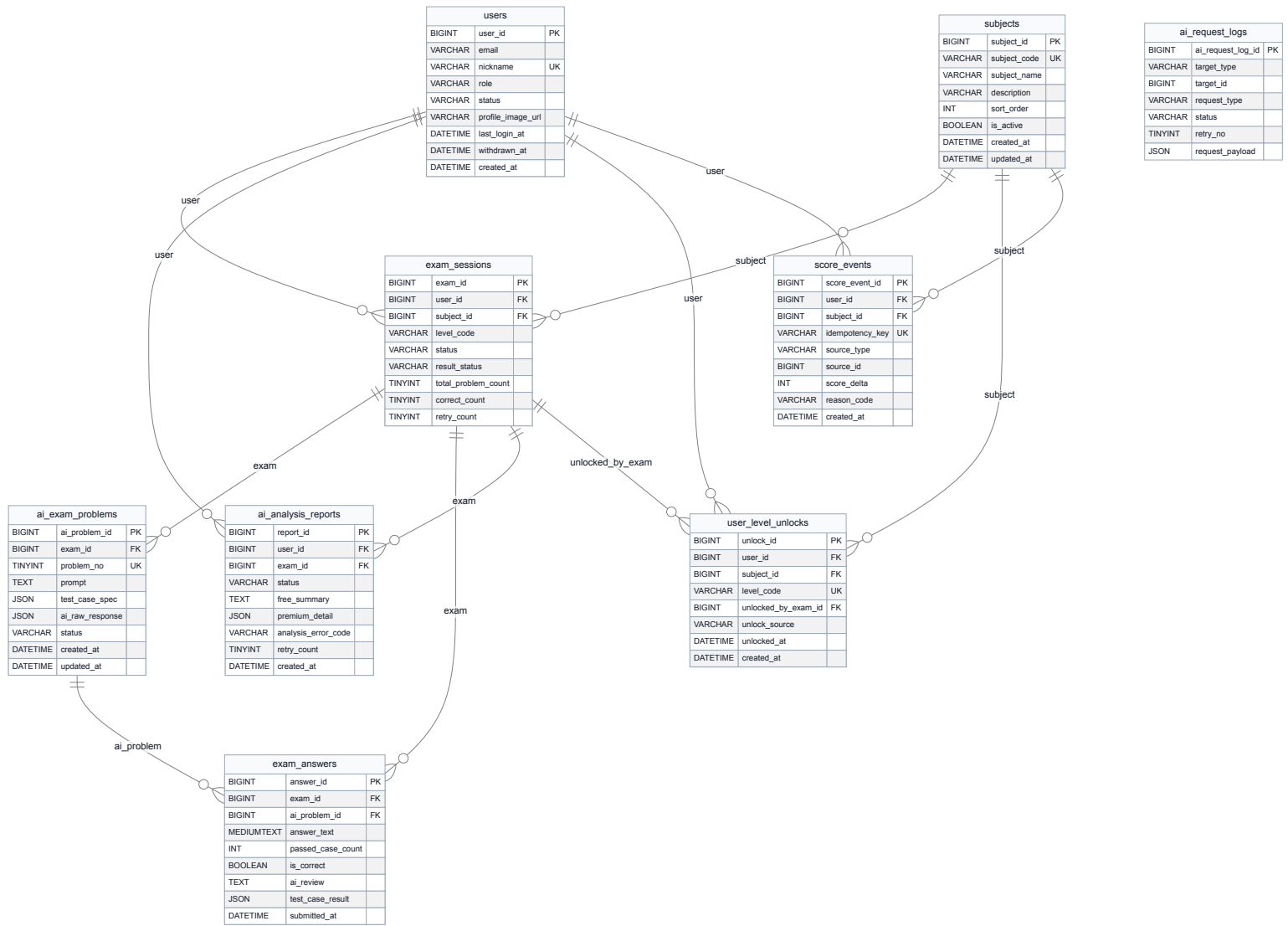
score_events

Column	Type	Key	NN
score_event_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
source_type	VARCHAR(30) NOT NULL		NN
source_id	BIGINT UNSIGNED NOT NULL		NN
score_delta	INT NOT NULL		NN
reason_code	VARCHAR(30) NOT NULL		NN
idempotency_key	VARCHAR(191) NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

ranking_scores

Column	Type	Key	NN
ranking_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NULL	FK	NULL
scope_type	VARCHAR(30) NOT NULL	UK	NN
scope_key	VARCHAR(100) NOT NULL	UK	NN
period_type	VARCHAR(30) NOT NULL	UK	NN
period_key	VARCHAR(30) NOT NULL	UK	NN
score	INT UNSIGNED NOT NULL DEFAULT 0		NN
rank_no	INT UNSIGNED NULL		NULL
calculated_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

AI 시험 / AI 분석



AI 시험 / AI 분석 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
exam_sessions	user_id	users	user_id
exam_sessions	subject_id	subjects	subject_id
ai_exam_problems	exam_id	exam_sessions	exam_id
exam_answers	exam_id	exam_sessions	exam_id
exam_answers	ai_problem_id	ai_exam_problems	ai_problem_id
ai_analysis_reports	user_id	users	user_id
ai_analysis_reports	exam_id	exam_sessions	exam_id
user_level_unlocks	user_id	users	user_id
user_level_unlocks	subject_id	subjects	subject_id
user_level_unlocks	unlocked_by_exam_id	exam_sessions	exam_id
score_events	user_id	users	user_id
score_events	subject_id	subjects	subject_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

exam_sessions

Column	Type	Key	NN
exam_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
level_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'CREATED'		NN
result_status	VARCHAR(30) NULL		NULL
total_problem_count	TINYINT UNSIGNED NOT NULL DEFAULT 3		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
retry_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
started_at	DATETIME NULL		NULL
submitted_at	DATETIME NULL		NULL
graded_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_exam_problems

Column	Type	Key	NN
ai_problem_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
problem_no	TINYINT UNSIGNED NOT NULL	UK	NN
prompt	TEXT NOT NULL		NN
test_case_spec	JSON NULL		NULL
ai_raw_response	JSON NULL		NULL
status	VARCHAR(30) NOT NULL DEFAULT 'GENERATED'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

exam_answers

Column	Type	Key	NN
answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
ai_problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
answer_text	MEDIUMTEXT NOT NULL		NN
passed_case_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
ai_review	TEXT NULL		NULL
test_case_result	JSON NULL		NULL
submitted_at	DATETIME NULL		NULL
graded_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_analysis_reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK	NN
status	VARCHAR(30) NOT NULL DEFAULT 'PENDING'		NN
free_summary	TEXT NULL		NULL
premium_detail	JSON NULL		NULL
analysis_error_code	VARCHAR(100) NULL		NULL
retry_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_request_logs

Column	Type	Key	NN
ai_request_log_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
request_type	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL		NN
retry_no	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
request_payload	JSON NULL		NULL
response_payload	JSON NULL		NULL
error_code	VARCHAR(100) NULL		NULL
error_message	VARCHAR(1000) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

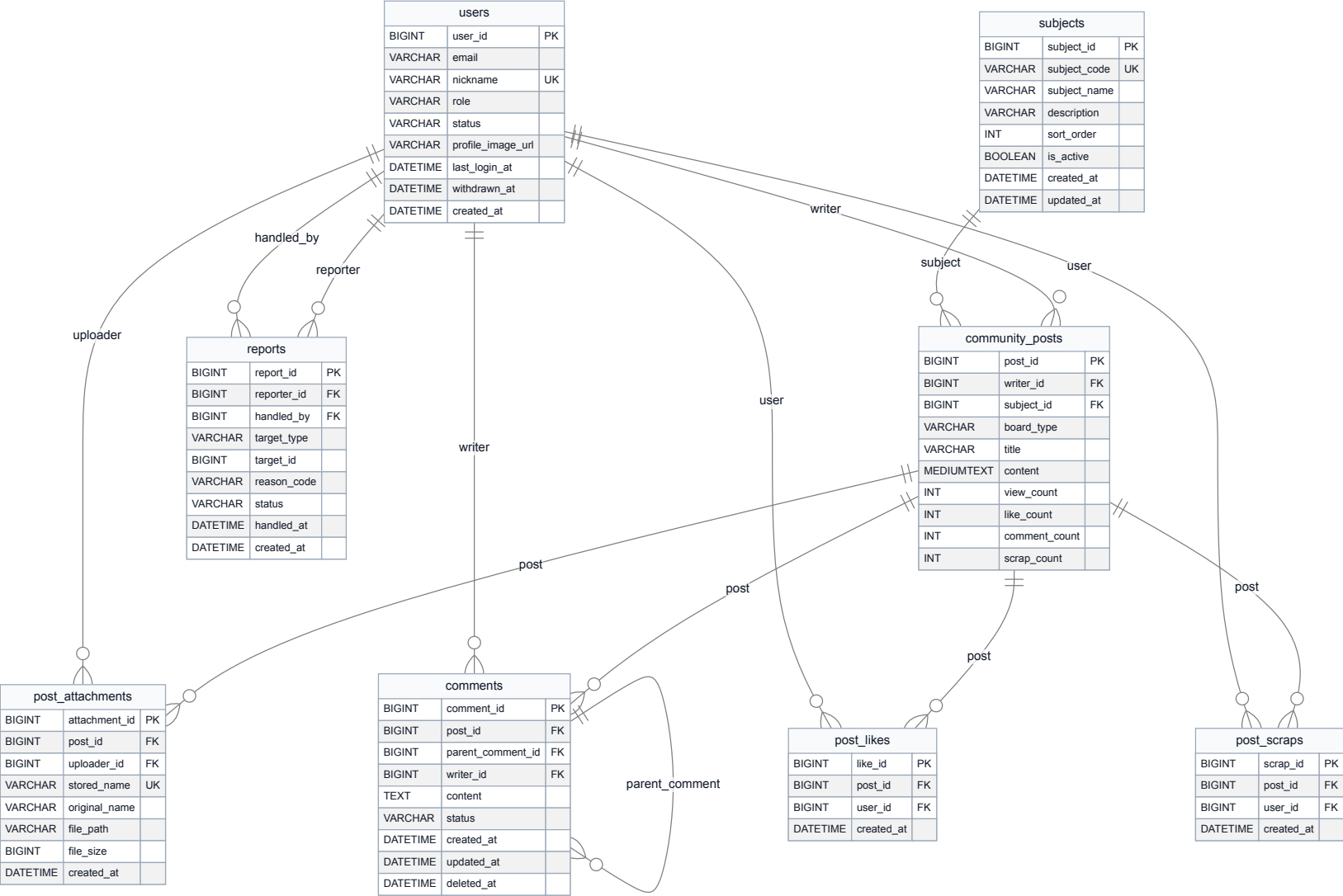
user_level_unlocks

Column	Type	Key	NN
unlock_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
level_code	VARCHAR(30) NOT NULL	UK	NN
unlock_source	VARCHAR(30) NOT NULL		NN
unlocked_by_exam_id	BIGINT UNSIGNED NULL	FK	NULL
unlocked_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

score_events

Column	Type	Key	NN
score_event_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
source_type	VARCHAR(30) NOT NULL		NN
source_id	BIGINT UNSIGNED NOT NULL		NN
score_delta	INT NOT NULL		NN
reason_code	VARCHAR(30) NOT NULL		NN
idempotency_key	VARCHAR(191) NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

커뮤니티



커뮤니티 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
community_posts	writer_id	users	user_id
community_posts	subject_id	subjects	subject_id
post_attachments	post_id	community_posts	post_id
post_attachments	uploader_id	users	user_id
comments	post_id	community_posts	post_id
comments	parent_comment_id	comments	comment_id
comments	writer_id	users	user_id
post_likes	post_id	community_posts	post_id
post_likes	user_id	users	user_id
post_scraps	post_id	community_posts	post_id
post_scraps	user_id	users	user_id
reports	reporter_id	users	user_id
reports	handled_by	users	user_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

community_posts			
Column	Type	Key	NN
post_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
board_type	VARCHAR(30) NOT NULL		NN
title	VARCHAR(200) NOT NULL		NN
content	MEDIUMTEXT NOT NULL		NN
view_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
like_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
comment_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
scrap_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL
deleted_at	DATETIME NULL		NULL

post_attachments

Column	Type	Key	NN
attachment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK	NN
uploader_id	BIGINT UNSIGNED NOT NULL	FK	NN
original_name	VARCHAR(255) NOT NULL		NN
stored_name	VARCHAR(255) NOT NULL	UK	NN
file_path	VARCHAR(500) NOT NULL		NN
file_size	BIGINT UNSIGNED NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

post_scraps

Column	Type	Key	NN
scrap_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

comments

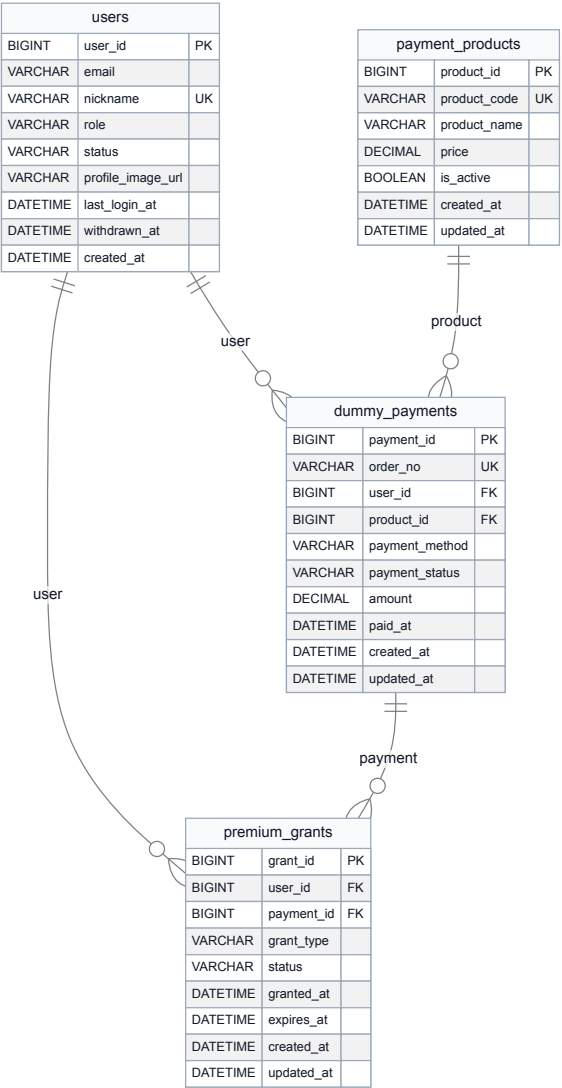
Column	Type	Key	NN
comment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_comment_id	BIGINT UNSIGNED NULL	FK	NU LL
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
content	TEXT NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL
deleted_at	DATETIME NULL		NU LL

reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
reporter_id	BIGINT UNSIGNED NOT NULL	FK	NN
reason_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'RECEIVED'		NN
handled_by	BIGINT UNSIGNED NULL	FK	NU LL
handled_at	DATETIME NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

post_likes

Column	Type	Key	NN
like_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN



결제 / Premium - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
dummy_payments	user_id	users	user_id
dummy_payments	product_id	payment_products	product_id
premium_grants	user_id	users	user_id
premium_grants	payment_id	dummy_payments	payment_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

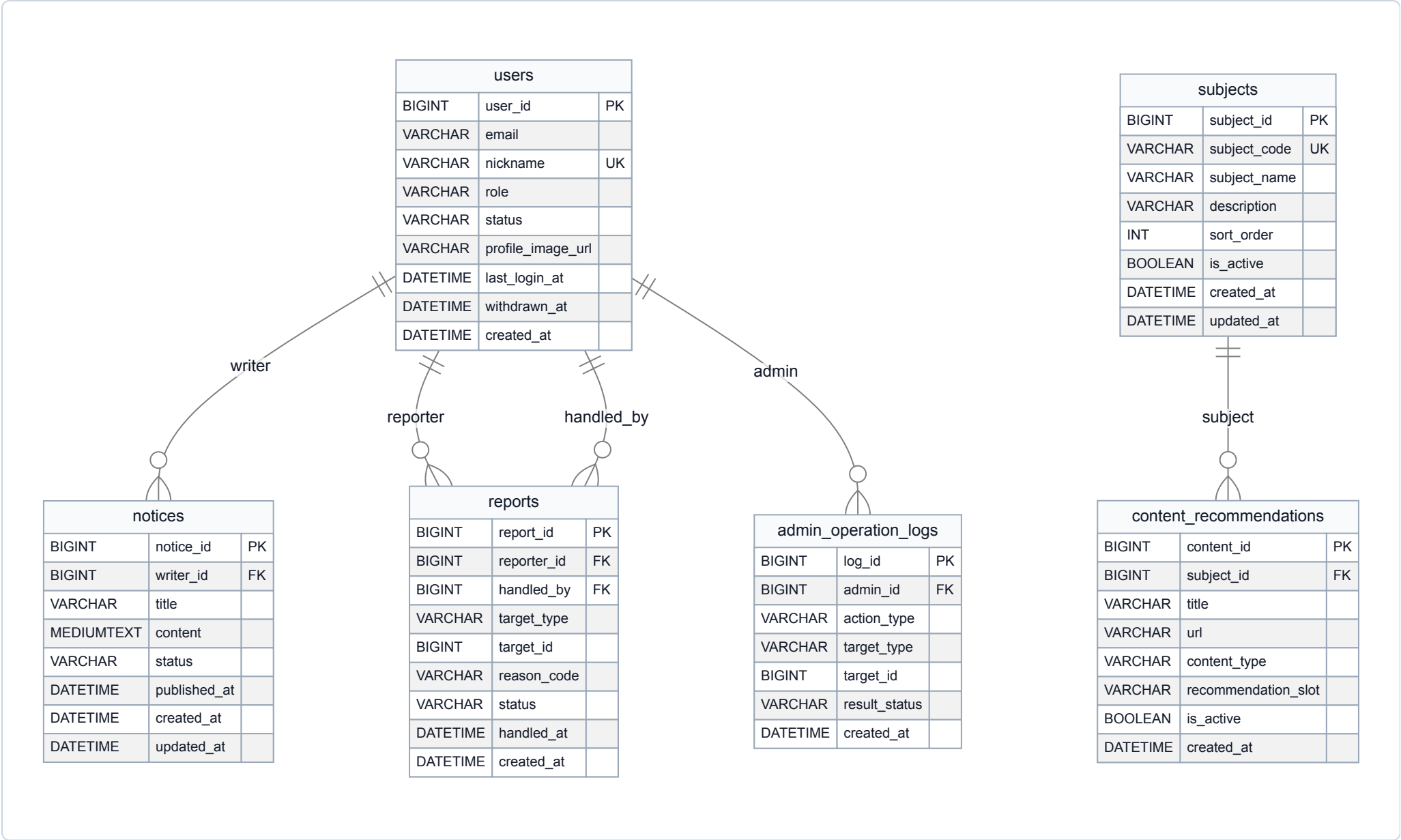
payment_products			
Column	Type	Key	NN
product_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
product_code	VARCHAR(30) NOT NULL	UK	NN
product_name	VARCHAR(100) NOT NULL		NN
price	DECIMAL(10,2) NOT NULL DEFAULT 0.00		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

dummy_payments			
Column	Type	Key	NN
payment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
order_no	VARCHAR(100) NOT NULL	UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
product_id	BIGINT UNSIGNED NOT NULL	FK	NN
payment_method	VARCHAR(30) NOT NULL DEFAULT 'DUMMY'		NN
payment_status	VARCHAR(30) NOT NULL DEFAULT 'READY'		NN
amount	DECIMAL(10,2) NOT NULL DEFAULT 0.00		NN
paid_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

premium_grants

Column	Type	Key	NN
grant_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
payment_id	BIGINT UNSIGNED NULL	FK UK	NU LL
grant_type	VARCHAR(30) NOT NULL DEFAULT 'LIFETIME'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
granted_at	DATETIME NOT NULL		NN
expires_at	DATETIME NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

관리자 / 공지 / 확장 콘텐츠



Child Table	FK Column	Parent Table	Parent PK
notices	writer_id	users	user_id
reports	reporter_id	users	user_id
reports	handled_by	users	user_id
admin_operation_logs	admin_id	users	user_id
content_recommendations	subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

notices			
Column	Type	Key	NN
notice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(200) NOT NULL		NN
content	MEDIUMTEXT NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'DRAFT'		NN
published_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
reporter_id	BIGINT UNSIGNED NOT NULL	FK	NN
reason_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'RECEIVED'		NN
handled_by	BIGINT UNSIGNED NULL	FK	NULL
handled_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

admin_operation_logs

Column	Type	Key	NN
log_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
admin_id	BIGINT UNSIGNED NOT NULL	FK	NN
action_type	VARCHAR(50) NOT NULL		NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NULL		NULL
result_status	VARCHAR(30) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

content_recommendations

Column	Type	Key	NN
content_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(200) NOT NULL		NN
url	VARCHAR(500) NOT NULL		NN
content_type	VARCHAR(30) NOT NULL		NN
recommendation_slot	VARCHAR(30) NOT NULL		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

v1.9 변경 요약

커뮤니티 조회수 및 seed 실행 안전성

- `community_posts.view_count`: 게시글 상세 조회 시 증가하며 목록의 조회수 정렬과 화면 표시에 사용.
- `Knowva_sample_data.sql`: DB v1.9 필수 컬럼 `preflight`와 단일 `transaction`을 적용해 오래된 `schema`에서 부분 반영되는 문제를 차단.
- `Knowva_DDL.sql`: 기존 데이터를 보존하는 `migration`이 아니라 전체 재생성 `script`임을 명시.

v1.8 변경 요약

인증 수단별 독립 계정 정책

- `users.email`: UNIQUE 제거. 이메일은 연락처/표시 정보이며 계정 병합 기준으로 사용하지 않음.
- `user_credentials.login_email`: 이메일/비밀번호 로그인 식별자. 이메일 가입은 인증 완료 후 별도 계정으로 생성.
- `social_accounts.provider + provider_user_id`: Google/GitHub 소셜 로그인 식별자. `provider` 이메일이 기존 회원과 같아도 새 `users` 계정 생성.
- `provider`: `google`, `github` 소문자 값으로 통일.

v1.5 변경 요약

v1.4 AI 정책 반영

- `ai_exam_problems.test_case_spec`: ChatGPT API가 생성/보조한 테스트케이스 후보 저장
- `exam_answers.passed_case_count`, `test_case_result`: 정답/오답 판정은 테스트케이스 실행 결과로 처리
- `exam_answers.ai_review`: AI 해설/코드 리뷰 저장
- `ai_request_logs`: ChatGPT API 요청/응답/실패/재시도 `payload` 기록

구분	반영 내용
일반 문제 세트	<code>practice_set_attempts.subject_id</code> 추가. 문제 세트 <code>attempt</code> 에서 과목을 직접 추적.
오답 복습	<code>wrong_answers.set_attempt_id</code> 추가. 오답이 발생한 문제 세트 <code>attempt</code> 를 직접 추적.
로그인/권한	관리자는 사용자와 같은 로그인 화면을 사용하고, 로그인 후 <code>role</code> 에 따라 학습 또는 관리자 메인으로 이동.
AI Provider	로컬 LLM 미사용. ChatGPT API 호출/응답/실패/재시도 기록을 <code>ai_request_logs</code> 에 저장.
AI 코딩테스트	AI는 문제 생성, 테스트케이스 생성 보조, 해설 생성, 코드 리뷰 담당. 정답/오답은 <code>testcase</code> 실행 결과로 판정.
DDL 실행	전체 실행 시 <code>elearning schema</code> 의 <code>Knowva table</code> 을 <code>DROP TABLE IF EXISTS</code> 후 다시 생성.